

Estimating Compensation Failure: How Much Has Compensation Been Underprovided?

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November 6 2025

1 Introduction

Understanding the distributional consequences of free trade has been important to economics and political science since at least Stolper-Samuelson ([1941](#)) and Rogowski's ([1987](#)) use of their theorem to demonstrate how the creation of gainers and losers affects politics. This importance has grown recently, with the seminal work of Autor, Dorn, and Hanson ([2013](#)) (henceforth ADH) suggesting at the magnitude of the losses from free trade and a large amount of evidence in political science showing that economic insecurity contributed to the rise of far-right populism (Scheiring et al. [2024](#)).

Consequently, there has also been a growing interest in understanding how much the losers have been compensated since theory suggests that compensation should mitigate the deleterious effects of free trade. Several prominent scholars have claimed that there has been a failure of compensation (Frieden [2019](#); Rodrik [2018](#)), with ADH ([2013](#); [2016](#)) themselves noting that trade adjustment assistance (TAA) in the U.S. has been limited.

However, these claims of compensation failure are unsatisfactory for two reasons. Firstly, it is unclear in absolute terms how large the losses from free trade really are. Many estimates, like those produced by ADH, do not take into account changes in prices and thus cannot pinpoint overall changes in real income. Secondly, and relatedly, there has been little attempt to compare observed compensation to the losses from free trade. Existing work focuses instead on changes in compensation, such as estimating how responsive TAA and other government transfers are to increased import exposure (Kim and Pelc [2021](#); ADH [2013](#)).

These measures, while illuminating, cannot answer the question of how much compensation has been underprovided, if at all.

In this paper, I provide the first estimates of compensation failure to my knowledge. Depending on how compensation is measured, I estimate that U.S. commuting zones that lost from free trade in the years 2000 and 2007 have only been compensated for 34-43% of their losses. My analysis is currently limited to these years because I build on the foundational work of Galle, Rodríguez-Clare, and Yi (2023) (henceforth GRY), who use data on these years and general equilibrium analysis to solve the first problem of estimating changes in real income from freer trade. My estimates suggest that TAA spending would have to have been more than doubled in the U.S. to compensate the losers of free trade.

The rest of this paper proceeds in three steps. Section 2 discusses the gains from trade as estimated by GRY. In section 3 I describe how I extend GRY to include a simple model of compensation. I present my estimates of compensation failure in section 4 and reintroduce GRY's gains from trade after adjusting for compensation. Section 5 briefly discusses limitations of the current analysis.

2 GRY: Welfare Gains and Losses from Free Trade

GRY develop a model of the economy that allows for multiple countries, multiple sectors, heterogeneous workers across sectors, and groups that can differ in their sector level efficiency or comparative advantage. While the authors develop several versions of the model in their paper, I adopt their baseline model in full. This includes, for example, their initial focus on tradable final goods and their preferred degree of worker heterogeneity across sectors ($\kappa = 1.5$). It is important to note that the estimated gains from trade do differ across versions of the model and that the baseline model has relevant limitations. For example, it does not allow for the mobility of workers across groups (i.e., geographic mobility) or allow for inequality in real income within groups. I refer readers to GRY for a more detailed

description of their model.

Due to current data limitations, I also follow GRY in only considering group level gains from trade for U.S. commuting zones in 2000 and 2007. However, while the authors devote much of their analysis to reconsidering the distributional effects of the China shock, I focus on a trade shock that returns the economy to autarky. I do this because while the authors are interested in addressing the findings of ADH, I am interested in the distributional effects from trade in general. The authors consider a similar autarky shock for the year 2000 (Figure A.9 on p. 369), so here I only present the results for 2000 to make my findings more comparable to theirs.

GRY define the welfare of workers in group g (i.e., commuting zones [CZs]) as their real income, denoted W_g . Before a trade shock, individual welfare in the group is simply average group income weighted by sector level prices.¹ After a trade shock, which I follow GRY in assuming is an infinite increase in trade costs for all sectors (i.e., a move to autarky), individual welfare is given by real income in the post-shock equilibrium, W'_g . GRY focus on estimating proportional changes in welfare, which are given by $\hat{W}_g \equiv W'_g/W_g$.² These changes in real income provide changes in utility and thus welfare under a social welfare function with no aversion to inequality (see section 2.7 starting on p. 341). Thus, the gains from trade GT_g are defined as the negative of $\hat{W}_g$ following an autarky shock ($100(1 - \hat{W}_g)$) and are interpreted as how much proportional income g would be willing to give up to make the shock not occur. Groups with proportional changes greater than one are estimated to have gained from trade (i.e., they would be worse off under autarky) while those with proportional changes less than 1 are estimated to have lost (i.e., they would be better off under autarky). **Figure 1** presents the authors' estimates for the gains from trade in 2000 (their Figure A.9, p. 369).

¹This is given as $(Y_g/L_g) \prod_s P_s^{\beta_s}$ in section 2.3 on page 337. Note that this implies that there is no within group inequality.

²GRY do not estimate W'_g but instead directly estimate $\hat{W}_g$ using equation (12) on page 337. However, note that $W'_g \equiv \hat{W}_g W_g$.

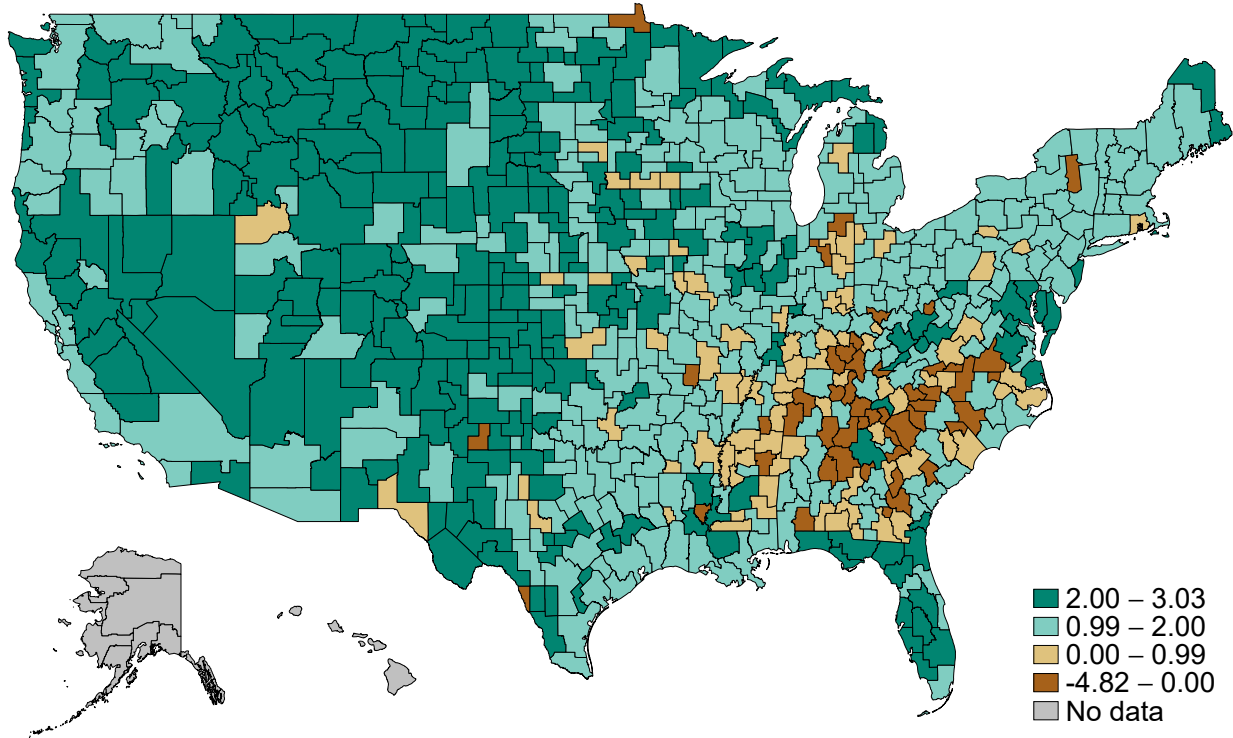


Figure 1: The authors' original estimates for the (negative) gains from trade after an autarky shock in 2000. I have modified the color scale to more clearly reflect gainers (commuting zones with proportional changes in welfare greater than or equal to 1) and losers (those below 1).

3 A Model of Compensation

In this section I explain how I extend GRY to model compensation. Firstly, I focus on absolute changes in welfare rather than proportional changes to compute total gains and total losses. Let $\Delta W_g = W'_g - W_g$ be the absolute change in welfare for group g from an autarky shock. The sum of the losses can be written as $\sum \nu = N$ where the set ν contains the absolute changes that are less than 0 (i.e., losses): $\nu = \{\Delta W_g \text{ s.t. } \Delta W_g < 0\}$. Likewise, the sum of the gains can be written as $\sum \eta = H$ where $\eta = \{\Delta W_g \text{ s.t. } \Delta W_g > 0\}$.

The compensation principle states that a policy is socially desirable if a hypothetical redistribution of income after the policy change can result in at least one household increasing its utility while no households lose utility (Moore 2007). Thus, full compensation for the

losers of free trade is the income needed to be redistributed from the gainers to the losers such that the latter do not experience a loss in real income. Here I consider a simple (regressive), lump sum tax and transfer system that results in full compensation. Specifically, a tax v on the gains must satisfy: $N + vH = 0$. In other words, the sum of the losses must equal the sum of the taxed gains for the tax to result in full compensation.³ This implies that the tax rate for full compensation must equal $v = -N/H$.

Let C represent the compensation parameter, which dictates how much of this tax occurs. Then the losers from free trade recoup a portion of their losses equal to C while the gainers keep only a portion of their gains equal to $1 - Cv$. Group welfare after the autarky shock and compensation can then be written as:

$$W_g'' = \{W_g' + Cv, W_g + (1 - Cv)\eta\}. \quad (1)$$

The proportional gains can now be written as $\hat{W}_g \equiv W_g''/W_g$.⁴ When $C = 0$, this model simplifies to GRY. When $C = 1$, full compensation occurs where the losers recoup their losses and the gainers keep the untaxed portion of their gains. **Figure 2** shows what the gains from trade displayed in **Figure 1** would look like under full compensation. Based on the estimates from GRY, v is computed to be 0.29. This implies that in 2000, a tax of 29% on the gains from trade was required to fully compensate the losers. As can be seen, after hypothetical compensation there are no commuting zones that lose while all the commuting zones that gained before compensation continue to gain. However, the maximum gain is now smaller, reflecting taxation. The histogram in **Figure 3** shows more clearly how GRY's estimated distributional effects from free trade change under full compensation.

³For the present I ignore the possibility of deadweight losses.

⁴Thus far I have not solved for $\hat{W}_g$ in equilibrium to account for the change in prices that would occur from a redistribution of income.

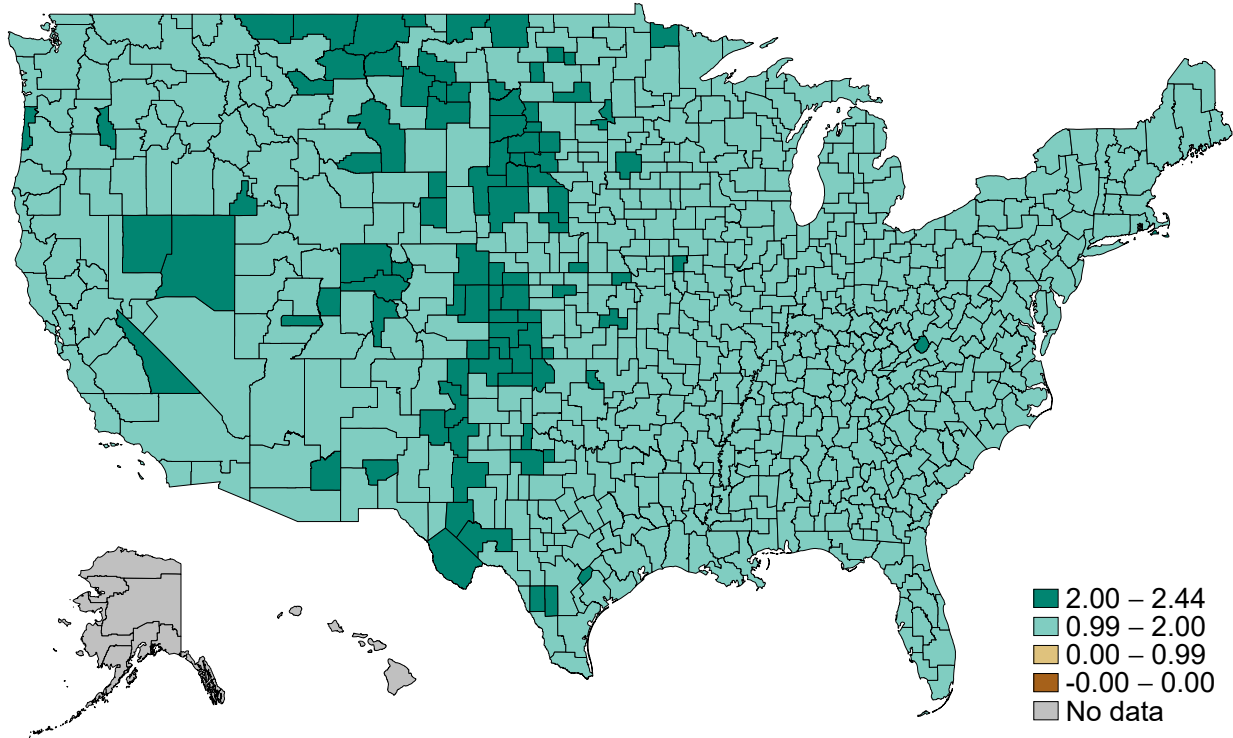


Figure 2: The gains from trade after an autarky shock in 2000 with full, hypothetical compensation (i.e., $C = 1$). Based on the estimates from GRY, full compensation would have required a 29% tax on gains.

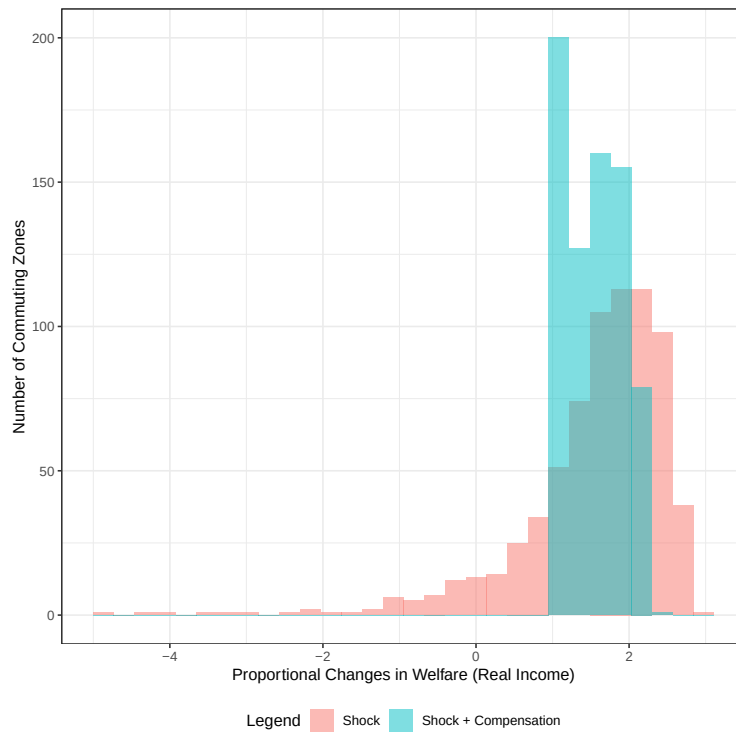


Figure 3: A histogram of GRY's gains from trade following an autarky shock before (red) and after compensation (blue).

4 Estimated Compensation Failure

The goal of this section is to describe how the above model can be used to estimate compensation failure and present initial estimates. Very simply, if $C\nu = C(W'_g - W_g)$ for $g \in \nu$ is the amount recouped by commuting zones that lost from free trade, then observed compensation equals $W'_g - W_g$ up to the true compensation parameter. Therefore, given data on compensation, $\hat{\beta}$ from the following regression would provide an estimate of C :

$$\text{Observed Compensation}_g = \alpha + \beta(W'_g - W_g) + e_g. \quad (2)$$

I use Kim and Pelc's (2021) measures of TAA spending at the commuting zone level as my data on observed compensation. Their replication files contain two measures, one where they assign state level TAA allocations to commuting zones based on the number of TAA certified workers and the other assigning state level TAA allocations to commuting zones based on unemployment claims (i.e., similar to ADH's commuting zone TAA measure). Because these measures and the estimated losses provided by GRY are in different units, I standardized both the TAA spending measures and the estimated losses. For the regression, I averaged across the estimated losses from 2000 and 2007 to match the single observation Kim and Pelc provide per commuting zone.⁵

Table 1 shows the results of the regression from equation (2), with $\hat{\beta}$ ranging from 0.34 to 0.43. Figure 4 updates Figure 1 after adjusting for compensation using the more generous estimate of 0.43 for C . These estimates imply that the losers in 2000 recouped between 0.34-0.43% of their lost income from free trade. Whereas it would have required a 29% tax on gains to recoup the losses, the more generous estimate of C implies that the observed tax rate was only $0.43 \times 0.29 \approx 0.12\%$.

⁵It is important to note that Kim and Pelc appear to average over 1990, 2000, and 2007 to create these TAA spending measures and do not provide the original figures. However, these regression results are similar if I only use the estimated losses from 2000, which suggests that averaging over an additional year may not disrupt this relationship.

	TAA (Workers)	TAA (Unemployment)
Intercept	0.000 (0.076)	-0.000 (0.079)
Losses	0.429 *** (0.076)	0.337 *** (0.080)
N	142	142
R2	0.184	0.114

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 1: Estimates of C using data on observed compensation from Kim and Pelc (2020).

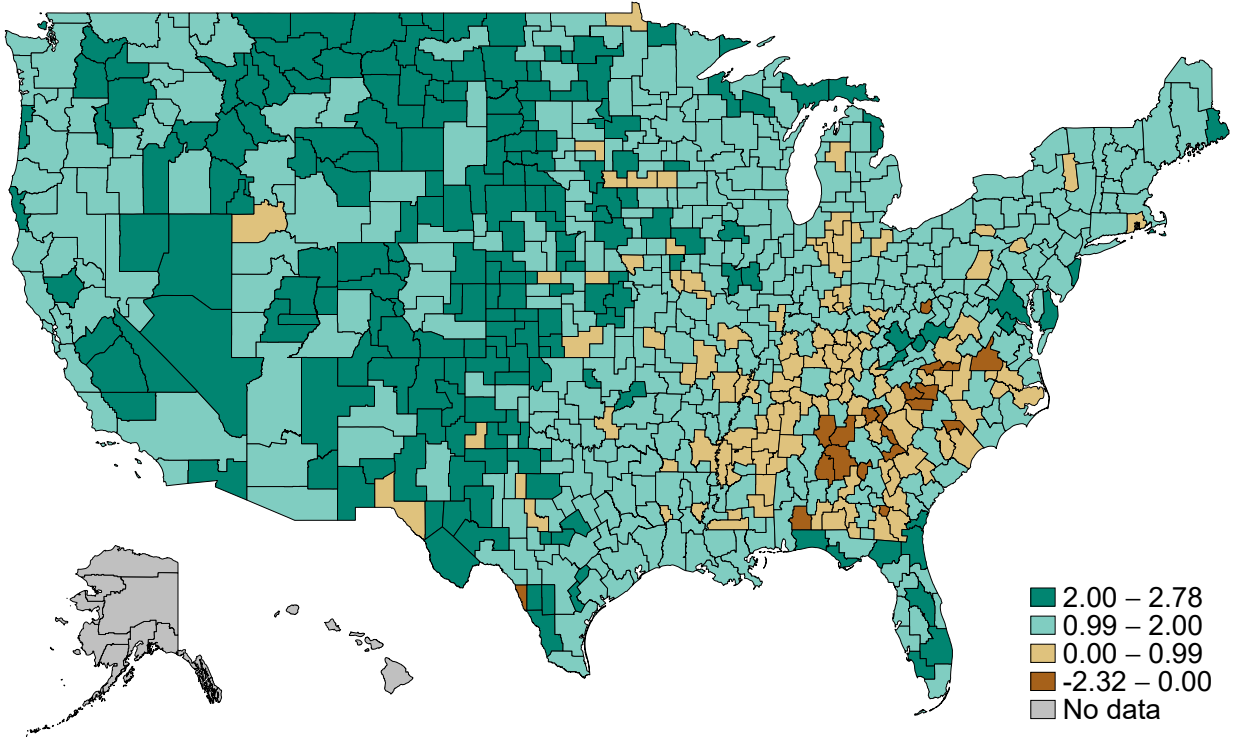


Figure 4: The estimated gains from trade after an autarky shock in 2000, adjusted for compensation using an estimate of 0.43 for C from Table 1.

5 Limitations

There are (at least) two limitations to the current analysis. The first is that there are very strong assumptions required for these figures to be taken at face value, especially about the utility of individuals and thus their welfare. If the losses from free trade were not strictly material, it is again unclear how much compensation was required for a full recoup of losses. However, these estimates may provide an upper limit in the sense that additional, immaterial losses would imply the need for greater redistribution to compensate the equivalent loss in utility. Thus, to the degree that the losses are immaterial compensation failure is being underestimated.

The second is that I have not modeled the deadweight losses from the tax required for compensation. This would act as a further tax on gains, which would again reduce the maximum gain observed after compensation. As noted in footnote 4, a related improvement on this analysis would take into account changes in prices from a redistribution of income (and from the deadweight losses) by solving for $\hat{W}_g$ in a new equilibrium. I intend to do this in the near future.

References

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